1:

We want to predict a fine-resolution relative poverty/vulnerability score for each spatial cell. That score should help us rank which places are more vulnerable than others, while still matching the trusted coarse official poverty totals when aggregated back up.

So the prediction task becomes: For each grid cell, predict a poverty-related score such that when we sum or average predictions across all cells inside a region, the result matches the official poverty value for that region. That is the core idea, a fine-scale prediction with coarse-scale supervision.

2:

Before merging anything, we need one shared spatial resolution. Since the datasets come at different resolutions, we should start coarser so we do not create fake precision. For example, we might choose a grid such as 2km cells for the Jamaica pilot. Every dataset will then be transformed so that each grid cell becomes one row in the modeling table.

Each row will represent:

- one spatial cell
- its location
- its input features
- the region it belongs to
- its population
- eventually its predicted poverty score

This gives us a consistent unit of analysis.

3:

This is the first major technical step. We take all the spatial datasets and align them to the same grid. For each cell, we collect all available features.

Possible features include:

- Relative Wealth Index
- settlement type/SMOD
- child population
- total population
- nighttime lights
- building density
- accessibility or distance-related features
- any other spatial proxy thought to correlate with deprivation

The merge happens by location. In practice, that means if data is raster-based, we overlay it on the target grid and aggregate or interpolate values into each cell, if data is polygon-based, we spatially join polygons to cells, if data is region-level, we assign the regional label to all cells inside that region

So after merging, the data table looks something like:

cell_id	region_id	x_coord	y_coord	RWI	SMOD	child_pop	lights	buildings	official_region_poverty

At this point, every cell knows:

- its own fine-scale features
- which coarse region it belongs to

- what official poverty number its region has

4:

This is the most important structural step. We do not have ground truth poverty for each cell. We only have ground truth for a larger region. So we connect each cell to its parent region using a region ID.

Example:
- Region A has official poverty = 0.34
- Region A contains 500 grid cells
- each of those 500 cells gets the region label “A”

Now, instead of supervising each individual cell directly, we supervise the model through aggregation:
- the model predicts a score for each of the 500 cells
- we aggregate those predictions over Region A
- we compare that aggregate to 0.34
- the difference becomes the training loss

So the model is trained indirectly. The case is not “this cell must equal 0.12” but “all cells in this region together must recover the official total”

That is what makes it a weakly supervised spatial learning setup.

5:

Now that the data is merged, each cell has a feature vector. For each grid cell, input might be:

$$x_i = [\text{RWI}, \text{SMOD}, \text{child population}, \text{nighttime lights}, \text{building density}, \dots]$$

The model takes that vector and outputs a scalar: $\hat{y}_i = f(x_i)$

Where $\hat{y}_i$ is the predicted poverty or vulnerability score for cell *i*.
At this stage, the model can be very simple:
- linear model
- small neural network
- XGBoost / LightGBM
- random forest

The notes are right: the exact model is not the main novelty at first. The main idea is the training setup and the aggregation logic.

6:

This is the core learning procedure. For each training region:
1. take all cells inside that region
2. run the model on each cell’s feature vector
3. get one predicted score per cell
4. aggregate those predictions into a regional prediction
5. compare that aggregated prediction to the known regional poverty total
6. compute the loss
7. update the model

If we use population-weighted aggregation, it would look conceptually like: $$\hat{Y}_r = \frac{\sum_{i \in r} \hat{y}_i \cdot p_i}{\sum_{i \in r} p_i}$$

where:

- r is a region
- pi is the population of cell iii
- yi is the predicted poverty score for that cell

Then the model minimizes the difference between:

- predicted region-level poverty Y(hat)r
- official region-level poverty Yr

So the loss could be something like: $$\mathcal{L} = \sum_r (\hat{Y}_r - Y_r)^2$$

This is how the model learns fine-scale structure without direct fine-scale labels.

7:

The model learns:

- which spatial features tend to be associated with higher poverty
- how those patterns vary within a region
- how to distribute a coarse poverty total across space in a way that is consistent with the features

For example, it may learn patterns like:

- lower wealth index + higher child population + rural settlement type = higher vulnerability
- certain combinations of building density and nighttime lights may mean different things in urban vs rural settings

Learning feature selection on its own depends on the model.

- Linear models do not really do feature selection unless regularized
- Lasso / regularized regression can shrink useless features
- XGBoost / LightGBM naturally prioritize useful features and largely ignore weak ones
- Neural networks learn complex feature interactions, but are less interpretable

Some models will implicitly learn which features matter more. But that does not mean we should blindly trust them. We should still do:

- feature importance analysis
- ablation studies
- correlation checks
- sanity checks by region type

8:

Start with Jamaica only.

- faster debugging
- easier to inspect maps
- easier to understand whether the method is working
- less complexity from cross-country variation

The Jamaica pilot should answer:

- can we successfully merge all datasets?
- can we train a model with region-level supervision?
- do predicted maps look reasonable?
- do aggregated predictions match official totals?
- do we beat a simple baseline like RWI?

This is the proof of concept.

9:

We do not want random cell-level splitting only, because nearby cells are spatially similar and that can make results look artificially good.

Instead, we want region-based splitting.

First level: hold out full regions

Example:

- 10 total regions
- train on 8
- test on 2 completely unseen regions

This tests:

- can the model generalize to new areas it has never seen?

Second level: hold out subregions within training regions

Inside the 8 training regions:

- hold out some smaller subareas or cells for validation/testing

This tests:

- can the model generalize locally within a familiar region?

So you get two forms of generalization:

1. within-region generalization
2. across-region generalization

That is much stronger than a simple random split.

10:

We should not jump straight to “our ML works.”

We need to compare it against simpler methods:

- uniform baseline: spread poverty evenly within a region
- RWI baseline: use wealth ranking directly
- heuristic baseline: current manual-style redistribution
- our ML model

This tells us whether ML is actually adding value.

If a simple RWI ranking performs just as well, then a complicated model is not justified.

11:

We need to test the model on more than one dimension.

A. Aggregation consistency

Does the model match official totals when predictions are aggregated back up?

B. Ranking quality

Does it correctly identify the most vulnerable cells or subregions?

Metrics could include:

- Spearman correlation
- top-k recall
- ranking overlap with known patterns

C. Generalization

Does it work on held-out regions?

D. Interpretability / trust

- Do the predicted maps look sensible?
- Can we explain what features the model is using?
- Can we identify where it is uncertain or likely wrong?

12:

Only after the Jamaica pilot is stable should we go broader.

Then the next step is:

- merge multiple countries or multiple larger regions
- keep some fully excluded for testing
- repeat the same training and evaluation framework

This lets us study:

- does the model generalize beyond Jamaica?
- do feature relationships transfer across different contexts?
- where does the method break?

That becomes the stronger publication-level result.